



Operator's manual for Attachment

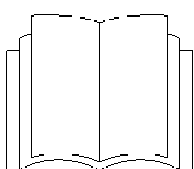


Auger drive unit

Direct drive unit and augers

for 200-series loaders: A21096

for 300-700-series loaders: A21100



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

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Appendix: EC Declaration of Conformity

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this AVANT attachment. It has been designed and manufactured on the basis of years of product development and experience. This operator's manual applies to all direct drive auger units manufactured and distributed by Avant that are listed on the cover page are covered by this manual. Other Auger models are also available from Avant, and the relevant manuals must be referred.

By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

The purpose of this manual is to help you to:

- operate the equipment in a safe and efficient manner
- observe and prevent any hazardous situations
- keep the equipment intact and ensure a long service life

This manual uses the following warning symbols to highlight factors that must be taken into account to reduce the risk of personal injury or property damage:

	Warning:
	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or property damage. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.



Note:

This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

With these instructions, even an inexperienced user can use the attachment and loader safely. The manual includes important instructions for experienced AVANT operators as well. Ensure that all persons using the loader have received proper guidance and familiarised themselves with the manual of the loader and attachment and all safety instructions before using the equipment. Keep this manual at hand throughout the service life of the equipment. If the manual is lost or damaged, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Tasks related to excavation may require a license before any work can be started. Contact your local authorities for more information about any authorisation needed before using the equipment. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. Original instructions.

2. Designed purpose of use

The AVANT Auger drive unit is an attachment designed and manufactured to be suitable for use with AVANT 200-700 series multi purpose loaders. The strong and robust hydraulically rotated drive unit is equipped with square shaft and pin auger locking system makes it possible to use different sized augers and to change them quickly. When used with small augers, the direct drive model is the fastest auger model for fence pole boring. The direct drive auger drive unit is best suitable when using smaller auger sizes in 100 mm to 300 mm range.

The auger blade bits are made of wear resistant hard tungsten alloy and they are replaceable for long service life. This makes it possible to use the augers even when drilling hard ground and the augers are well suited also for professional use. When operating with larger augers and hard ground, the planetary gear equipped HD drive units are recommended, which can provide more torque, that is needed to drive the augers. The larger augers work the best when operated on area where there are no larger stones. Augers and blade bits are available from your Avant retailer. The auger drive unit is intended for making upright or nearly upright holes into soil using suitable earth augers, extension shafts and accessories, while operating the auger and the controls of the loader from the driver's seat. Other use of the auger drive unit or the auger bits is prohibited.

The attachments have been designed to require as little maintenance as possible. The operator can perform regular maintenance tasks, but certain more demanding tasks, such as service or replacement of the hydraulic components, must be left to qualified professionals. Appropriate protective equipment must be used. Spare parts must be similar to original specifications, which can be ensured by using only original Avant spare parts. Auger blades and blade bits, as well as spare parts, are available from your Avant retailer.

Read the maintenance and servicing instructions in this manual. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you need spare parts or maintenance services.

3. Safety instructions for using an auger

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful and improper or careless use or maintenance may cause serious personal injury or property damage. Operating this attachment includes hazards that cannot be protected against with mechanical means or product design. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not completely familiarised yourself with its operation and the related hazards.



Misuse or careless use, or using an attachment that is in bad condition, may cause serious injuries. Familiarise yourself with the controls of the loader and coupling of the attachment at a safe area, especially how to stop the equipment in a safe manner. Read the following safety precautions carefully.

Read the safety instructions carefully before handling the attachment



- When attaching the attachment to the loader, **ensure that the locking pins of the loader's quick attach plate are in the lower position and that they have locked the attachment to the loader.** Also make sure that the auger tools are securely locked to the drive unit.
- Note that the drive unit is hinged and moves freely. When transporting, the drive unit must be tilted forward, so that the auger is supported by the supporting cradle of the mounting bracket. Do not tilt the auger up excessively, as it could tip over. Keep the auger pointed towards the ground in a position where the auger will not swing or hit the loader, ground, or any obstacle.
- The auger is intended to be used by one operator at a time and both the auger and the loader must be stopped before leaving the operator's seat. Stop the auger as instructed below in the safe stopping procedure before allowing others near the machine. Keep a minimum safety distance of at least 1 metre from an auger that is in operation and from raised boom.
- Going under a raised attachment or boom is dangerous; the boom may lower unexpectedly due to loss of stability, equipment failure, or other persons using the controls of the loader. Always lower the auger on the ground before leaving the driver's seat.
- Survey the operating area and plan work ahead. Make sure in advance that it is safe to dig at the operating area. Ensure the stability of the loader and firmness of ground also on even terrain.
- Never leave the driver's seat when the auger is rotating. Keep hands and feet away from rotating tools. Do not wear loose clothing or, for example, long scarves or jewellery. Also, cover long hair. Watch out for entanglement hazard around rotating tool.
- The loader will move slightly caused by the oscillating motion of rotating auger, make sure that moving will not cause hazards. Do not press the loader boom down too hard, and keep front tyres in contact with the ground.
- Always transport the auger drive unit as low and close to the machine as possible to keep the centre of gravity low and for the best stability. Avoid using the equipment on horizontally tilted terrain and exercise extreme caution when driving on rough or tilted ground, as the machine can tip over.
- When driving, pay attention to the contours of the terrain and, for example, loose rocks as well as other hazards, such as branches and trees reaching the driver's area. Pay attention to the surroundings and other persons and machines moving in the vicinity.
- Make sure that you use only an attachment that is in proper working condition. Perform daily inspections and read the instructions on service and maintenance. Never use the attachment if the hydraulic systems of the attachment and the loader are not completely in order.
- Make sure that the attachment is properly supported for maintenance and inspections. Store the auger properly and make sure that the auger is supported when removing its locking pin. Never perform any maintenance or adjustments, or change auger tools, while the loader engine is running.
- Do not modify the structure of the attachment in a manner that would affect its safety. Use only suitable earth auger drills. Other use is prohibited.
- Also read the safety instructions shown in the operator's manual of the loader. Familiarise yourself with the controls and safe operation of the loader at safe area.



Make sure that there are no electric cables, water lines or similar obstacles within the reach of the equipment. If unsure, contact the property owner, town services and i.e. local electric, water, and gas companies. Find out about local rules and requirements before any excavation work.



Note that the loader boom moves down and may be tilted even if the engine has been stopped. Never go under a raised boom or auger. Always lower the tip of the auger on the ground before leaving the driver's seat. Operate the controls of the loader only when sitting on the driver's seat.

Safe stopping of the equipment, before going near the attachment:

- Stop the attachment, turn auxiliary hydraulics control lever to its neutral position.
- Lower the boom and set the tip of the auger on the ground, or place the auger horizontally on the ground, lowering the boom completely.
- Make sure that the loader will remain stable and shut down the loader engine.
- Lock the parking brake.
- Release residual pressure from the hydraulic system.



Remove key from ignition and make sure that another person can't start the loader or move the loader boom, before going near the auger. If the equipment is not stopped safely, a slight movement of an auger that has become stuck may cause the auger to start again, causing an entanglement hazard.

Remember to wear proper protective clothing:



- The noise level in the driver's seat exceeds 85 dB(A). Wear hearing protection while working with the loader.



- Wear safety boots while working with the loader.
- Wear protective gloves. When handling hydraulic hoses or components, wear also safety goggles.
- Do not wear loose fitting clothing or, for example, long scarves or jewellery, which can get entangled with the auger. Also cover long hair.

4. Technical specifications

Table 1 – Auger drive unit specifications

Product number	A21096	A21100
Weight:	100 kg	
Maximum torque:	1355 Nm	
Suitable earth augers:	75-300 mm	
Auger fixing:	57 mm square	
Rotating speed:	0,4-1 /s (24-60 rpm)	0,4-2,4 /s (24-144 rpm)
	See picture 2, auger rotating speed	
Hydraulic energy input:	15 – 70 l/min max 22,5 MPa (225 bar)	
Suitable AVANT loader:	200 series	300-700 series

Picture 1 – Auger drive unit main dimensions

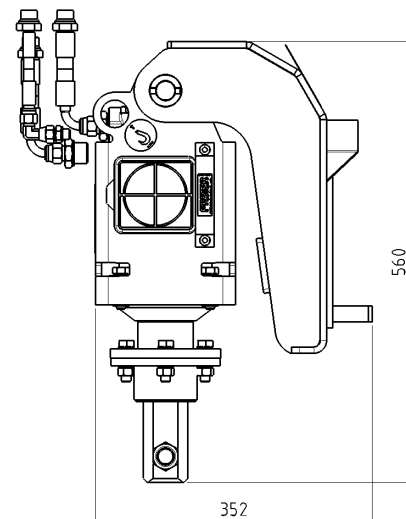


Table 2 – Auger information

Diameter	75 mm	100 mm	150 mm	200 mm	225 mm	250 mm	300 mm
Product number:	A34225	A33601	A33608	A33609	A33610	A33611	A33612
Drilling depth:	1500 mm	1500 mm	1500 mm	1500 mm	1500 mm	1500 mm	1500 mm

Auger wear part numbers:

Auger pilot, tungsten*	A32561	Auger pilot, earth-type	A32536
Blade teeth, tungsten*	A32560	Blade teeth, earth-type	A32535

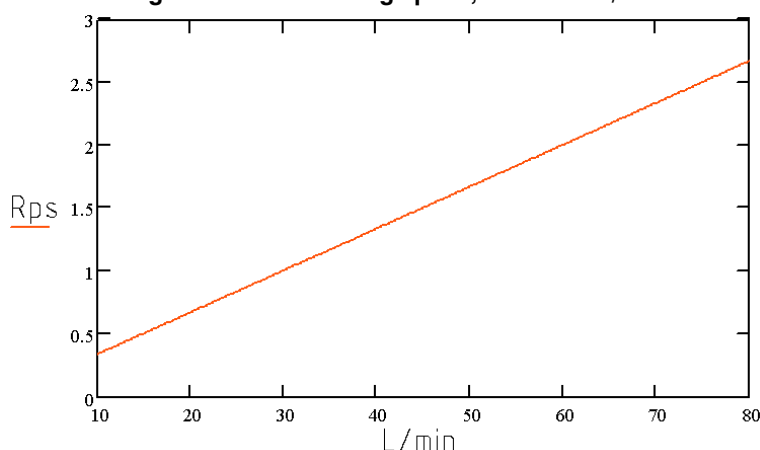
Up-to-date list of augers is available from your dealer or from Avant website at www.avanttecno.com

*As standard, all augers are equipped with tungsten wear parts.

Table 3 – Extension shafts

Length	500 mm	800 mm	1000 mm
Product number:	A33546	A32532	A33554

Picture 2 - Auger drive unit rotating speed, revolutions/s

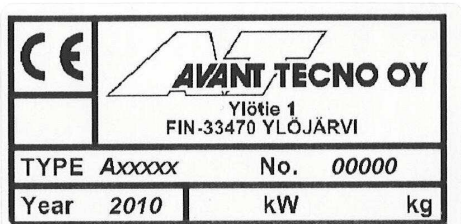


4.1. Safety labels

Listed below are the labels and markings, which must be visible on the equipment. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer or contact information provided on the cover.



The warning labels contain important safety information and they help to identify and remember the hazards related to the equipment. Replace damaged or missing warning labels with new ones.



Attachment identification plate



A46771



A46803



A46772



A46806

Table 4 – Decal warning messages and locations

Decal	Explanation
A46771	Read instructions before use.
A46803	Crushing hazard, keep away from hinged drive unit.
A46772	Do not go under a raised attachment or the loader boom, keep a safe distance from lifted equipment.
A46806	Beware of entanglement around the auger; do not go near the equipment while it is rotating. Watch out also entanglement of loose clothes, hair, jewellery or similar. Keep a safe distance and follow safe stopping procedure.
	Identification plate of the attachment

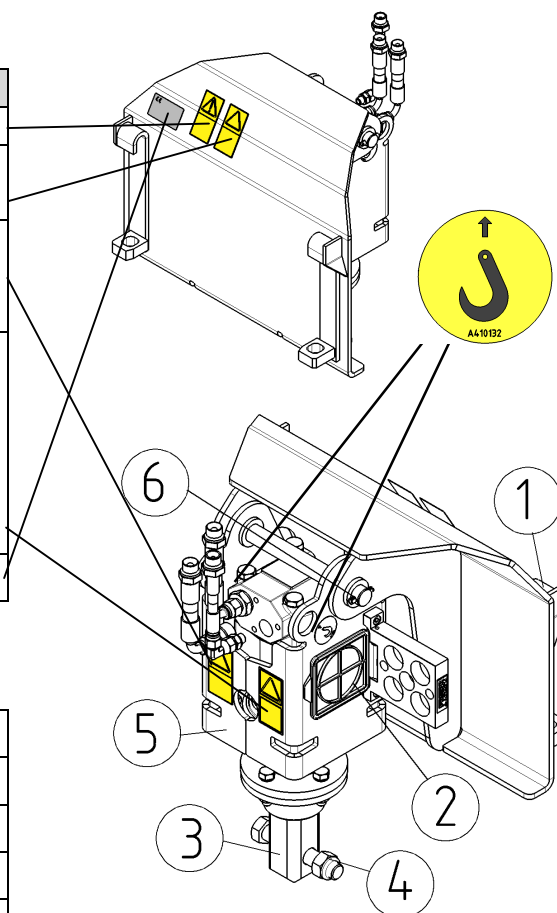


Table 5 – Main components of auger drive unit

1	Quick coupling plate
2	Multiconnector parking stand
3	Tool shaft, square
4	Auger locking pin
5	Drive motor cover
6	Swinging joint

5. Assembly of the auger

The attachment is ready for use as delivered. Fit the attachment and its hydraulic hoses to the loader as instructed below.

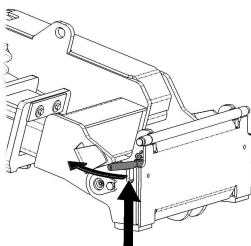
5.1. Coupling the auger to the loader

Attaching the attachment to the loader is quick and easy, but it must be done carefully. The attachment is mounted to the loader boom by using the quick attach plate and the counterpart on the attachment. If the attachment is not locked to the loader, it may detach from the loader and cause hazardous situations. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions shown in Chapter 3 and additional information shown in the operator's manual of the loader. The attachment is mounted to the loader as follows:

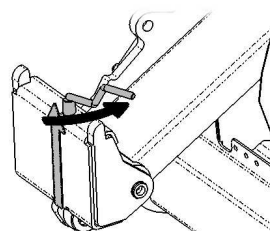
300-700 series loaders:

Step 1:

- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- If your loader is equipped with a hydraulic attachment locking system, see additional instructions about the use of the locking system from the relevant manuals.
- Ensure that the hydraulic hoses are not in the way during installation.

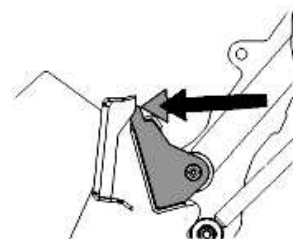
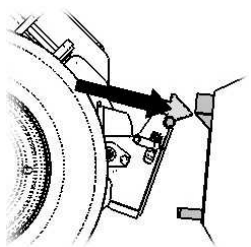


200 series loaders:



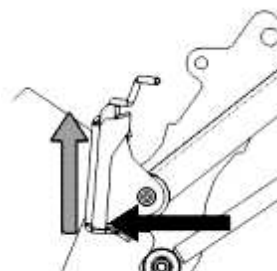
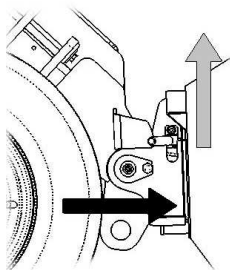
Step 2:

- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this. Align the upper pins of the loader's quick coupling plate so that they are under the corresponding brackets of the attachment.
- Depending on the storage position of auger drive unit and the reach capability of the loader, it may be necessary to assist the mounting of the attachment manually. See Chapter 5.3 for correct uncoupling procedure and storage position.



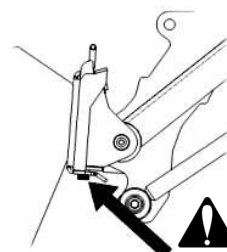
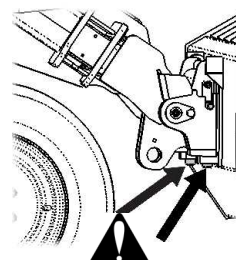
Step 3:

- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lock the locking pins manually or lock the hydraulic locking. Always check the locking of pin / pins.





An attachment that has not been completely locked to the loader may fall on the boom or operator, or under the loader during driving, and cause serious injuries. Before moving or lifting the attachment, make sure that the locking pins are in the lower position and come through the fasteners on both sides. Never move or lift an attachment that has not been locked.



Do not lift an unlocked auger drive unit or use the automatic locking of the locking pins at a height of over one metre. The hinge joint of the auger drive unit may cause the drive unit to tip over. It is recommended to lower the attachment on the ground and secure the locking manually, as it is anyway necessary to leave the driver's seat to connect the hydraulic hoses.

5.2. Connecting and disconnecting the hydraulic fittings

On 300-700 series loaders hydraulic hoses are connected using the multiconnector system. . If you want to change the type of the hydraulic couplings, contact your Avant dealer or service point for instructions or installation services. On 200-series loaders only conventional quick couplings are available.



Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure or if the control lever of the auxiliary hydraulics control lever is locked on. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead to unintended movements of the attachment, or ejection of high-pressure fluid, which can cause serious injuries or burns.



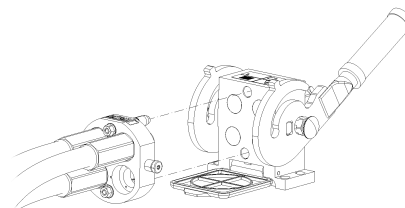
When uncoupling the attachment from the loader, always disconnect the hydraulic fittings first before unlocking the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent dirt from entering the hydraulic system.

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place them on top of the equipment.

Connecting the multiconnector system:

Align the pins of the attachment connector with corresponding holes of the loader connector. The multiconnector will not connect if the attachment connector is upside down. Connect and lock the multiconnector by turning the lever towards the loader.

The lever should move easily all the way to its locking position. If the lever does not slide smoothly, check the alignment and position of the connector and clean the connectors. Also shut down the loader and release the residual hydraulic pressure.



To disconnect the multiconnector system, turn off the auxiliary hydraulics of the loader, and unlock the multiconnector. To unlock, push the knob on the side of the connector, and turn the lever to disconnect.

Conventional quick couplings (alternative couplings for some loader models):

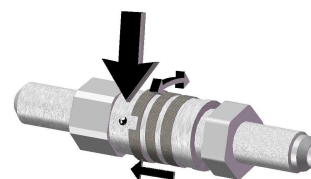
Before connecting or disconnecting the standard quick couplings, the residual pressure must be released as shown below. The conventional quick couplings will not connect, if there is pressure in the hydraulic system.

To connect and disconnect the standard couplings, move the collar at the end of the female fitting. The hoses should be connected so that the fitting equipped with a coloured cap is connected to the corresponding fitting of the loader. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. When disconnecting the standard quick couplings a small amount of oil may drip from the couplings. Wear protective gloves and have some cloth at hand to keep the equipment clean.



Make sure that all three couplings are properly connected, otherwise the hydraulic motor of the mower may get damaged.

In a 200-series machine, the hose can be connected by pushing the fitting in the counterpart. When connecting and disconnecting the 200-series hydraulics quick couplings, you should pay attention to their locking. There is a small groove in the collar of all female fittings, and a small ball near the collar. When connecting or disconnecting, the groove and ball must be aligned. The locking prevents the female and male fittings from being disconnected from each other by accident when the attachment moves. The collar may also turn accidentally to the locking position, making it impossible to connect or disconnect the couplings.



Before disconnecting conventional quick couplings shut down the loader and move the auxiliary hydraulics control lever of the loader to its extreme positions to release the residual pressure of the hydraulic system.

Releasing residual hydraulic pressure:

To make sure that there will not be residual pressure in the hydraulic system of the attachment, shut down the loader engine and move the auxiliary hydraulics control lever of the loader to its extreme positions before disconnecting the couplings. If the fittings will not connect to the loader, the residual pressure must be released. The residual pressure must be released before handling the blades of the attachment and before storage of the attachment.

5.3. Uncoupling the attachment and storage

The drive unit is heavy (around 100 kg), so careful uncoupling to a position that can be reached with the loader is important. Remove dirt from the auger and clean the equipment before storage. Refer to maintenance instructions in chapter 7

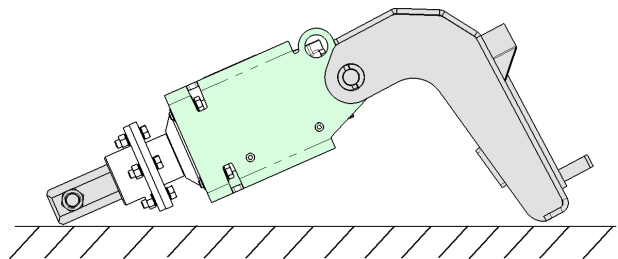


Avoid storing the attachment or its tools directly against the ground for long periods to avoid damage from corrosion. See chapter 7 for maintenance instructions before long storage periods.

Correct storage position and uncoupling procedure:

For the easiest coupling and uncoupling of the attachment, augers and extension shafts should be removed from the drive unit. Leaving the auger tool on the drive unit may cause bending of the tool and rust during longer storage.

Place the shaft end of the drive unit on a pallet, and drive in reverse motion with the loader and turn the boom control lever simultaneously, so that the auger will be in position shown in the adjacent figure. Disconnect hydraulic fittings, unlock the locking pins and uncouple the attachment by operating the controls of the loader and reverse away with the loader.



Loader that is not equipped with a telescopic boom:

If the loader is not equipped with a telescopic boom may not reach the quick coupling plate of the drive unit if the drive unit is stored directly against the ground. In this case it may be necessary to manually change the position of the auger for better reach with the loader.

Drive the loader as close to the quick attach plate of the auger drive unit as possible and tilt the attachment plate of the loader forward to minimise lifting height. If possible, lift and tilt the quick attach plate of the attachment to similar position as shown above, and place blocks of wood under the quick attach plate to make it possible to reach the attachment normally.

5.4. Transporting

Always stop the auger before driving with the loader. The auger drive unit should point forward so, that it rests on the quick coupling frame of the attachment. Always transport the auger as low and close to the loader as possible for the best stability. Drive cautiously and avoid swinging of the drive unit, or hitting the ground. When stopping, always lower the auger or the drive unit on the ground.

6. Instructions for use

Check the attachment and the operating environment once more before starting to work. Quick inspection of the equipment and the operating area before use are parts of ensuring safety and the best performance of the equipment.



Keep a safe distance to a rotating auger and stay back if other person operates the controls of the loader. Do not grasp the auger or try to control or guide it manually. Beware of entanglement hazard; wear only tight fitting clothing.

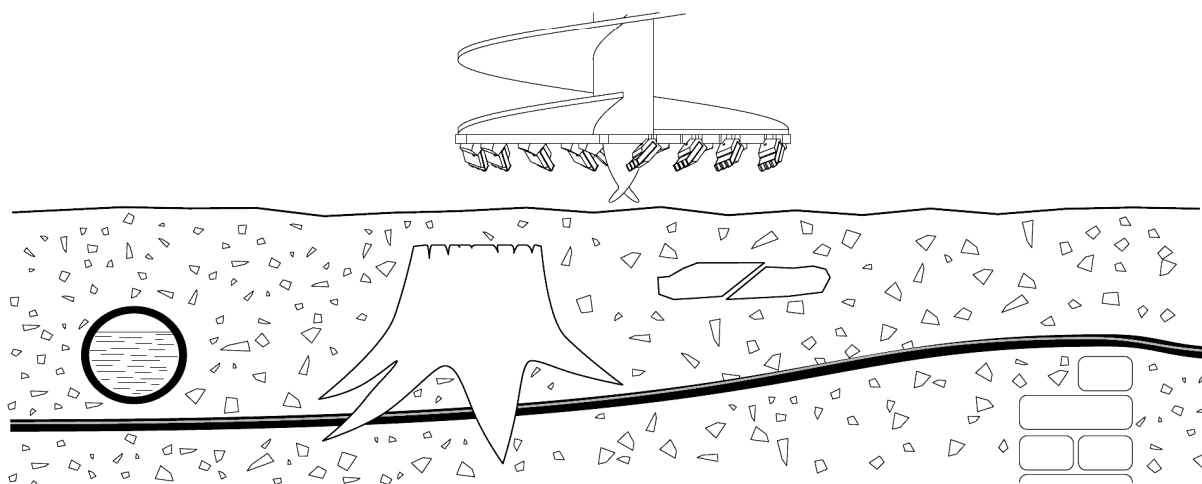


6.1. Checks before use

- Check the general condition of the attachment and the loader, and check for possible hydraulic oil leaks. The attachment must not be used, if there is a fault in the hydraulic system of the loader or the attachment. Check that the drive unit is intact, it can rotate freely to correct operating direction and that there is no string or other foreign material coiled around the auger. Refer to Chapter 7 for maintenance instructions.
- Check the condition of the auger bits as shown in Chapter 7.
- Ensure that the loader and the attachment are used in a safe and intended manner. Do not allow children to operate the equipment.
- Operate the attachment from the driver's seat only. Remember correct working methods and avoid leaving the driver's seat unnecessarily.



Plan the drilling operation in advance. Make sure that is safe to dig at the area. Explore the operating area and find out about possible electricity cables, water mains or other possible obstacles before starting work. Hitting an obstacle may cause a serious accident. If in doubt about possible obstacles, seek professional advice and consider using detection equipment. You may be required by law to contact your local authorities before using an auger to make sure that there are no hidden and potentially dangerous obstacles in the ground.





Overheating of hydraulic oil decreases the power and affects the service life of the hydraulic components of both the attachment and the loader. Clean the loader oil cooler core regularly; refer to the loader manual for more instructions.

6.2. Operating the auger



Start the auger only when beginning to use it. Do not leave the equipment running unattended. Do not go near the equipment when another person operates the controls of the loader.

6.2.1. Fitting an auger tool and using extension shafts

The auger tools can be fitted when the drive unit is locked to the loader, so that the position of the auger can be controlled with the loader. The auger tools should be fitted after arriving at the operating area, to lower the possibility of uncontrolled swinging of the drive unit while driving.

To fit an auger tool:

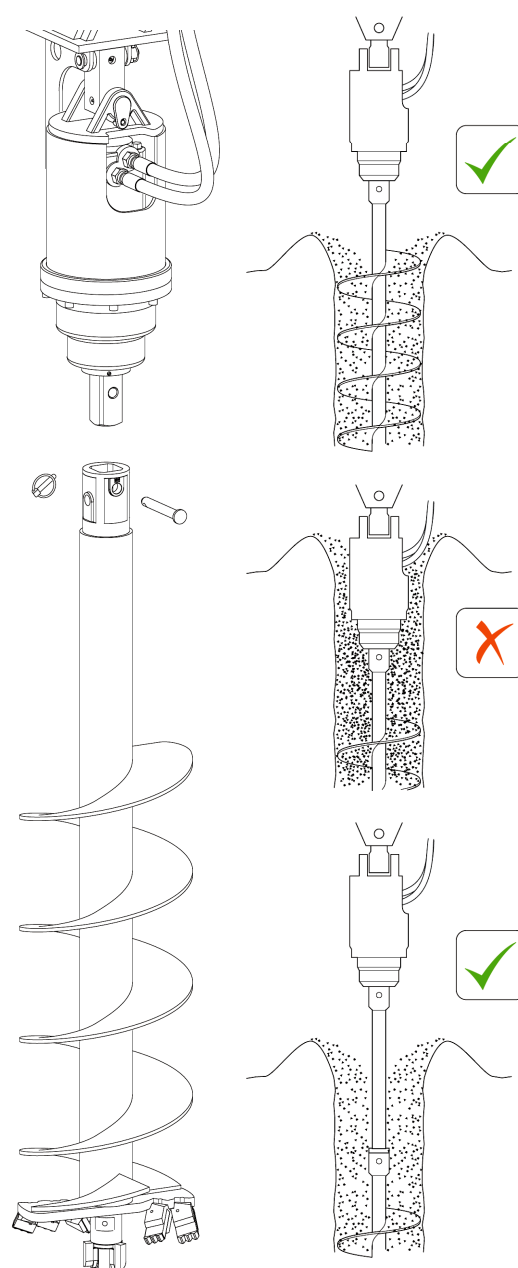
Clean the contact surfaces of the auger tool and the drive unit before fitting. The augers can be fitted in vertical or horizontal position. Especially the larger augers should be placed in vertical position due to their weight.

1. Clean the contact surfaces of the auger and the drive unit.
2. The easiest way to fit the auger depends on the size and weight of the auger. Depending on the size and height of the auger, the tools should be fitted in vertical position, allowing lowering the auger drive shaft into the auger tool. Smaller and lighter augers can be fitted when the drive unit is in a tilted position. Align the holes for the locking early enough.
3. Always fit and secure the locking pin or locking screw.

Using extension shafts:

Do not drill deeper than the reach of the auger. Drilling beyond the reach of the auger can lead to difficulties, if the auger becomes stuck or gets damaged. Using extensions allows drilling deeper holes, but maximum depth of 2,5 metres must be observed. Extension shafts must be fitted only when needed at the work site.

Fit the extension sufficiently in advance for easier handling, and so that the fixing point remains above ground surface. When fitting the extensions after reaching previous maximum depth, care must be taken so that the auger will not fall deeper into the hole.



6.2.2. Operating the auger



Entanglement around the auger or getting crushed under the loader boom can cause serious injuries or death. Do not leave the driver's seat while the auger is in operation or when it is lifted off from the ground. Also, keep others at a safe distance from the equipment and stop the auger if there are others closer than 1 meter from the auger. Follow safe stopping procedure when going near the auger is necessary or changing tools.

The auger drive unit is started by turning the auxiliary hydraulics control lever of the auger to its locking position. Start the auger at a safe area to see the rotating direction; make sure that the auger rotates **clockwise**. Set the engine rotating speed according to operating conditions. Operating at full rotating speed may be possible, but generally modest speed will give best results.

Position the auger against the ground in upright position. Gradually lower the boom to apply force on the auger. The needed force to push the auger down depends on the operating conditions and type of soil. Also, the bigger the auger, the more time should be allowed for the auger to lift soil before operating the loader controls. Do not push down too hard, so that the front wheels of the loader remain in contact with the ground to avoid stalling the auger. Also, make sure that the quick coupling plate is not in contact with the auger drive unit, so that the hinge will allow the auger to move.



Note that in addition to controlling the loader boom, the position of the auger must also be corrected by using the drive pedals or the telescopic boom. Raise the auger out of the ground regularly to clear material from the auger and to maintain drilling effectiveness. Do not try to clean the auger by swinging it with force or by hitting it, it may bend or its teeth may get damaged from impact.

Raise the auger out of the ground regularly to clear material from the auger. Smaller augers may need manual cleaning especially when working on moist soil. Frequent clearing of the hole will also make it easier to pull the auger out from the hole. Keep the auger rotating the same direction when pulling out, so that material will also be pulled.

Augers can't pull or break large stones from the ground. The augers are not intended for other purposes making holes on various types of soil. If penetration speed decreases clearly, raise the auger and check for stones.



Keep the drilling position upright so that the auger will pull itself efficiently. Drilling at an angle may cause bending of the auger.



If the auger hits the ground, the blade bits may crack or other damage may result. Handle the augers with care, so that the blades remain in the best possible condition.



7. Maintenance and servicing

The attachment has been designed to be as maintenance free as possible. Maintenance includes cleaning and monitoring the general condition of the attachment and its tools.



Ensure that the drive unit is properly supported during all inspection and maintenance procedures. All maintenance procedures must be performed while the loader engine is shut down and the residual pressure is released and the hydraulic hoses are disconnected, and the boom has been lowered completely.

Spare parts are available from your nearest AVANT representative. Please Contact your dealer if you have any additional questions, or if you need any maintenance services.

7.1. Checking hydraulic components

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a leak in the hydraulic system of the attachment or the loader. Leaking hydraulic fluid may penetrate skin and cause serious injuries. Seek medical attention immediately in case hydraulic fluid penetrates the skin. Hydraulic fluid is also harmful to the environment. Operate only with hydraulic oil that is accepted for use in Avant loaders.



Never handle hydraulic components when the hydraulic system is pressurized, since a fitting may break or become loose and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.



Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of any hose has worn off. The hoses have limited service life, which is marked on the hoses. After the date marked on the hoses, they should be replaced with new ones. If there are signs of leaks, to check a component, hold up a piece of cardboard in the area where a leak is suspected.

Finding any of these faults means that the hydraulic hose or component must be replaced. Spare parts are available from your nearest AVANT retailer or service point. Leave the repair work to professional service technicians, if you don't have experience about hydraulic assemblies and how to perform the repairing safely.

7.2. Cleaning the equipment

Pressure washer and a mild detergent may be used to wash the auger. Spraying on hydraulic components or the labels on the attachment must be avoided. Clean the attachment before long storage periods and touch up paint where needed to prevent rust damage. Apply oil on the auger connecting surfaces, the locking pin and corresponding holes to ensure smooth operation and to prevent surface rusting.

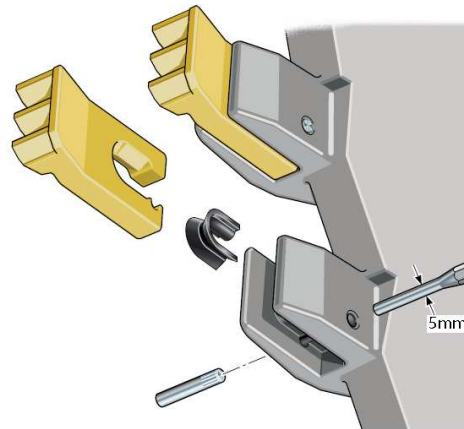
7.3. Auger blades

The hard-alloy tungsten tipped blades have good wear resistance capabilities. The performance of the auger decreases as the blades wear, or if the auger gets damaged. Worn teeth or pilot can drastically reduce auger penetration and reduce auger service life.

Check auger before each day of use for loose, worn or broken cutting teeth and pilot head. Hard operating conditions may require checking teeth and point at more frequent intervals. Replace the blade bits before the blade locking, fixing points or the auger get damaged. It is recommended to keep spare teeth and pilot head on hand. The hard-alloy blades can not be sharpened, they must be replaced when worn. New blades and other spare parts are available from your Avant dealer.

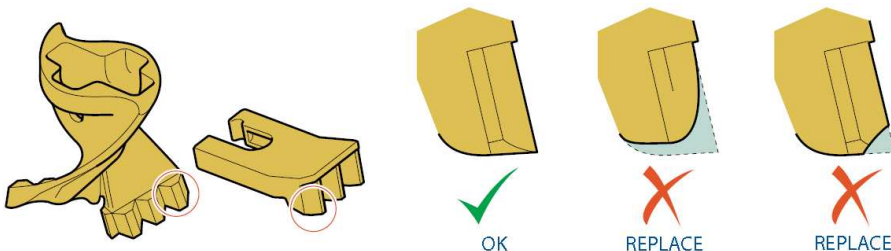
Shock lock teeth:

Use a 5 mm pin punch to drive the retaining pin out through the bottom of the tooth holder. The blade tooth and Shock Lock rubber can then be pulled out.



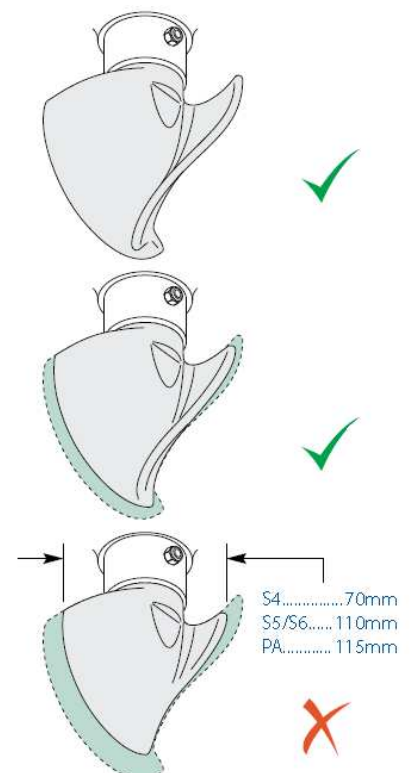
After setting a new rubber piece and a replacement Shock Lock tooth to the slot, insert a new retaining pin into the top of the tooth holder, plain end first. Drive the pin fully in, use a pin punch to make sure that the knurled end of the pin is fully engaged in the hole.

Shown below are the recommended replacing points for blade bits. Replace blades before the tooth holders or the auger gets damaged.



Pilot head:

Check the pilot head by measuring its wear as shown in the adjacent figure. Handle the augers carefully and avoid damaging the pilot head by making sure to transport it so that the head will not hit the ground.



8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) from the date of purchase for the attachment it manufactures.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
EG-försäkran om överensstämmelse
EC Declaration of Conformity
EG-Konformitätserklärung



Valmistaja / Tillverkare / Manufacturer / Hersteller:

AVANT TECNO OY

Osoite / Adress / Address / Adresse:

Ylötie 1

33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveystaamimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:
 SFS-EN ISO 12100, SFS-EN ISO 4413

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats:
 SFS-EN ISO 12100, SFS-EN ISO 4413

We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:
 SFS-EN ISO 12100, SFS-EN ISO 4413

Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt:
 SFS-EN ISO 12100, SFS-EN ISO 4413

Kuvaus: Avant-kuormainten kanssa käytettävä maaporan pyöritin.
 Beskrivning: Jordbör; arbetsredskap för Avant lastare.
 Description: Auger drive unit; attachment for Avant loaders.
 Beschreibung: Erdbohrer; Anbaugerät für Avant Radlader.

Mallit / Modeller / Models / Modellen:

Avant	200 series	300-700 series
Maapora Jordbör Auger Erdbohrer	A21096	A21100

Allekirjoitus / Namnteckning / Signature / Unterschrift:

22.12.2011 Ylöjärvi, Finland

Risto Käkelä,
 Toimitusjohtaja / Verkställande direktör /
 Managing Director / Geschäftsführer





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